

ECE4813 ARM SoC DESIGN

Mini-Project Description

Playing Card Recognition and Strategy Game using the Arm Based ECE4813-SoC

Platform	DE10-Nano: dual-core Arm Cortex-A9 HPS with Cyclone V FPGA fabric
Vision Input	Terasic camera connected through GPIO-based camera interface
Output	HDMI display with optional UART terminal for debugging
Team Size	Three students per team
Assessment Points	Week 6, Week 9 and Week 12

This document defines the technical scope, development requirements, milestone demonstrations, assessment rubrics and submission requirements for the ECE4813 mini-project.



1. Introduction: Playing Card Recognition and Strategy Game

The mini-project requires each team to develop an embedded vision and game-control system using the DE10-Nano platform. A Terasic camera observes playing cards placed in a defined playing area. The system must acquire the camera image, detect and classify card ranks or symbols, maintain the game state, and present the recognised cards and scores through an HDMI display.

The project progresses from a basic SoC demonstration to a memory-pair game and finally to an advanced King–Jack–Joker strategy challenge. The staged structure allows teams to demonstrate hardware–software integration before attempting real-time recognition and autonomous game logic.

1.1 Project objectives

- Design and integrate an Arm-based SoC using the hard processor system and FPGA fabric of the DE10-Nano.
- Use the two Arm Cortex-A9 CPU cores where appropriate, with an RTOS/LINUX on at least one processing context for advanced credit.
- Acquire images from the Terasic camera and detect playing-card ranks and symbols.
- Implement or accelerate a CNN-based recognition function in the FPGA fabric where feasible.
- Develop game-state logic for card matching, scoring, captures and recovery actions.
- Generate a clear HDMI user interface showing detected cards, board position, team status, scores and system state.
- Demonstrate individual technical understanding through viva questions.

1.2 Expected system behaviour

1. The camera is positioned to focus on one or more playing cards in the defined game area.
2. The SoC captures the image and performs preprocessing and card recognition.
3. The recognised card identity and relevant game status are presented on the HDMI display.
4. Switches or push-buttons may be used for start, reset, mode selection or user confirmation.
5. UART may be used as a bidirectional debugging and diagnostic interface.
6. The system stores card states, recognised positions, scores and game history in on-chip or external memory.

2. Development of the Arm Based ECE4813-SoC

Each team shall develop a customised SoC platform, referred to in this project as the Arm Based ECE4813-SoC. The implementation should combine the dual-core Arm Cortex-A9 hard processor system, FPGA-based logic, memory, communication interfaces and peripheral I/O required by the playing-card application.

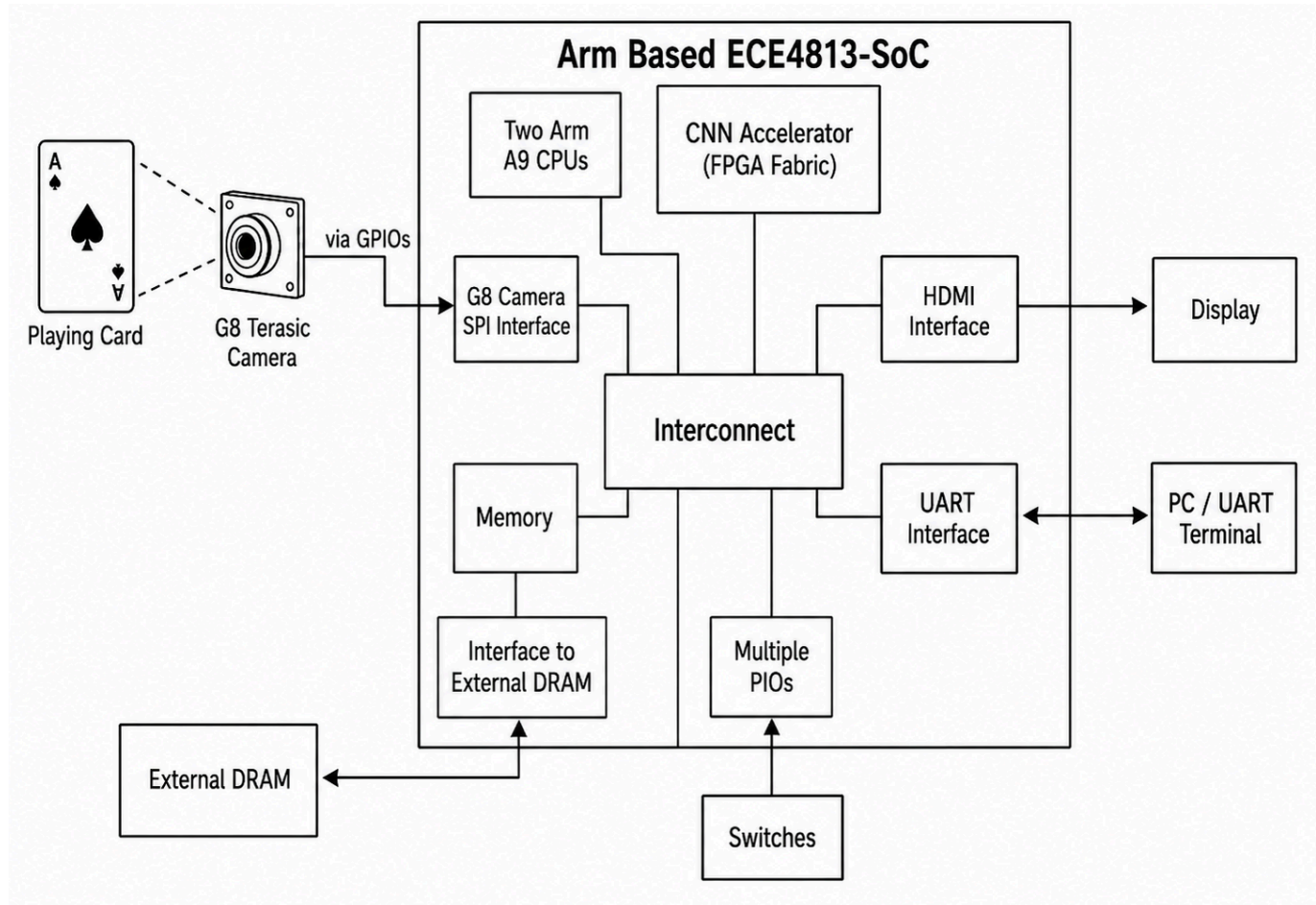


Figure 1. Proposed block diagram of the Arm Based ECE4813-SoC for playing-card recognition.

2.1 Minimum SoC features

Subsystem	Minimum expected function
Two Arm Cortex-A9 CPU cores	Application control, game logic, communication, display management and/or task partitioning.
System interconnect	Connect processors, FPGA accelerators, memories and memory-mapped peripherals.
CNN accelerator in FPGA fabric	Hardware-accelerated inference or selected CNN operations. A software CNN alone may be used initially, but the FPGA acceleration must be clearly identified when claimed.
Camera interface	Acquire frames or pixels from the Terasic camera through the board GPIO camera connection and associated control interface.
HDMI interface	Drive the display and present detected card identity, position, match result, score and game state.
UART interface	Bidirectional communication with a PC or serial terminal for diagnostics and system verification.
On-chip memory	Buffers, program/data storage, frame fragments, feature maps or game-state storage.
External DRAM interface	Bidirectional storage for frame buffers, software, CNN data and runtime information.
Multiple PIOs	Inputs from switches or push-buttons and optional outputs such as status LEDs.

2.2 Recommended hardware–software partitioning

Function	Suggested implementation
Image capture and pixel stream handling	FPGA fabric with DMA or frame-buffer support.
Image preprocessing	FPGA logic for resize, crop, grayscale, thresholding or normalization where practical.
CNN inference	CNN accelerator in FPGA fabric, or hybrid FPGA preprocessing with Arm-based inference.
Game logic and scoring	Arm processor software, preferably organised as RTOS/LINUX tasks.
HDMI graphics and overlays	FPGA display pipeline with software-controlled text, icons and status data.
System control and diagnostics	Arm software, UART terminal and PIO inputs.

2.3 Technical clarification: CNN implementation

The preferred technical term is “CNN accelerator implemented in the FPGA fabric” . The accelerator may implement the complete inference network or selected computationally intensive layers. Teams must state clearly which operations execute in the FPGA fabric and which execute on the Arm processors.

2.4 Team and individual responsibilities

The system may be developed collaboratively, but every student must understand the architecture and their assigned contribution. Suggested individual responsibility areas include:

- Camera acquisition and image preprocessing.
- CNN model, FPGA accelerator or recognition algorithm.
- Arm software, RTOS/LINUX task structure and inter-processor coordination.
- HDMI controller, graphics overlay and game dashboard.
- Game-state machine, scoring, capture and recovery logic.
- System integration, testing and performance measurement.

3. Milestones, Marking Scheme and Rubrics

The project will be assessed through three progressive demonstrations. Each milestone mark is provisional until the assessors cross-verify the contribution and understanding of individual team members through viva questions. Marks may be adjusted individually when a student cannot explain the design or implementation presented by the team.

Week	Milestone	Focus
Week 6	Milestone 1	Basic SoC build, multicore/RTOS/LINUX capability and initial card-symbol detection.
Week 9	Milestone 2	4 × 4 memory game with card recognition, matching and HDMI presentation.
Week 12	Milestone 3	Advanced King–Jack–Joker strategy game, captures, recovery and innovation.

3.1 Milestone 1 – Basic SoC Demonstration (Week 6)

The team shall demonstrate a functioning Arm Based ECE4813-SoC and establish the basic camera-to-display processing path. This milestone assesses SoC construction and early recognition capability rather than complete gameplay.

Criterion	Marks	Performance evidence
Basic functional SoC build and demonstration	50	The integrated system boots or runs reliably and demonstrates processor, FPGA, memory and at least one peripheral interface.
Use of both Arm Cortex-A9 cores, with an RTOS/LINUX used on at least one processing context	25	Clear task allocation, scheduling and evidence that both cores contribute meaningfully.
Basic CNN implementation	15	A CNN model or accelerator is integrated and produces a verifiable inference output.
Playing-card digit/number detection	5	The system detects and displays at least one numeric playing-card rank under controlled conditions.
Playing-card alphabet detection	5	The system detects and displays at least one alphabetic rank such as A, J, Q or K.
Total	100	Subject to individual viva verification.

3.1.1 Milestone 1 rubric

Achievement level	Descriptor
Excellent	All claimed features operate reliably; integration is clear; students explain architecture, interfaces and design decisions confidently.
Good	Most features operate with minor limitations; students understand the main design and can diagnose basic issues.
Satisfactory	Basic SoC demonstration works, but integration or recognition is limited and explanations lack depth.
Insufficient	The system is incomplete, unstable or largely prebuilt without evidence of team understanding.

3.2 Milestone 2 – Card Recognition and Memory Game (Week 9)

A 4 × 4 array of cards will be placed face down. Two competing teams take alternate turns. During each turn, two cards are opened and remain exposed for a maximum of five seconds. If the cards do not form a matching pair, they are returned face down at the same locations.

3.2.1 Game rules

- The system shall recognise both exposed cards and associate each recognised card with its board position.
- A correct pair remains matched or is recorded as collected according to the competition procedure.
- An unmatched pair is returned to its original positions after the exposure interval.
- The system must successfully identify matching pairs on at least two separate occasions; one isolated success is not sufficient evidence.
- The HDMI display should show card identity, position, turn, match result, score and remaining cards.

3.2.2 Marking scheme

Criterion	Marks	Performance evidence
Card recognition and display	70	Correct recognition of exposed cards, including rank/symbol and board position, with the result shown on the display.
Correct matching of pairs at least twice	20	The system identifies two or more valid pairs in separate attempts, demonstrating repeatable operation.
HDMI presentation quality	10	Clear, organised and informative display of detected cards, positions, score, turn and game progress.
Total	100	Subject to individual viva verification.

3.2.3 Milestone 2 rubric

Achievement level	Descriptor
Excellent	High recognition reliability, two or more confirmed matches, responsive game flow and a polished HDMI dashboard.
Good	Recognition and matching work in most trials; display is clear with minor omissions.
Satisfactory	Recognition works only under restricted conditions or matching is inconsistent; display presents basic information.
Insufficient	Cards are not recognised reliably, board position is not tracked, or matching is primarily manual.

Competition recognition: the winning team will receive one bonus mark toward the overall unit assessment and may be recognised through suitable certificates, subject to unit approval and assessment policy.

3.3 Milestone 3 – Advanced King–Jack–Joker Strategy Challenge (Week 12)

The final milestone introduces strategic game logic using four Kings, four Jacks and four Jokers arranged face down in random order. The cards selected in each turn are exposed for a maximum of five seconds. The King Team and Jack Team compete to collect the greatest number of Jokers.

3.3.1 Game rules

- King Team collects a Joker by opening one King and one Joker in the same turn.
- Jack Team collects a Joker by opening one Jack and one Joker in the same turn.
- If King Team opens a King and a Jack together, the Jack is captured. If Jack Team opens a Jack and a King together, the King is captured.
- A team may return one previously collected Joker to the central pool to recover its captured King or Jack. The recovered card returns to play, and the team loses that Joker.
- The team collecting the greatest number of Jokers wins.
- The HDMI display must indicate detected cards, Joker totals, captured cards, active team, recovery actions and the current game state.

3.3.2 Marking scheme

Criterion	Marks	Performance evidence
Card detection and recognition	50	Reliable detection of Kings, Jacks and Jokers, including card identity and location.
Correct HDMI indication	20	Accurate display of recognised cards, team status, Joker count, captured cards and game progress.
Successful capture of three Jokers	15	The game logic records and displays three valid Joker captures through the required King/Joker or Jack/Joker combinations.
Successful capture of an opponent card	10	The system correctly implements and displays the King–Jack capture rule.
Viva and individual understanding	5	Each student explains their contribution, architecture, algorithm and test evidence.
Total	100	Before innovation bonus.

3.3.3 Innovation bonus

A bonus of up to 10 marks may be awarded for an extraordinary additional game or intelligent extension that clearly demonstrates capabilities of the developed SoC beyond the prescribed requirements. The innovation must be functional, technically explained and distinct from the core milestone requirements. The final unit-level treatment of bonus marks remains subject to the approved assessment policy.

3.3.4 Milestone 3 rubric

Achievement level	Descriptor
Excellent	Reliable recognition, autonomous game logic, three valid Joker captures, opponent capture, recovery mechanism, clear HDMI status and strong viva performance.
Good	Core game logic and display operate with minor restrictions; most capture actions are demonstrated correctly.
Satisfactory	Card detection works, but strategic actions require assistance or are not consistently tracked.
Insufficient	The final game is largely manual, recognition is unreliable, or the team cannot explain the implementation.

4. Assessment Conditions and Demonstration Requirements

- All demonstrations must use the team’s Arm Based ECE4813-SoC implementation on the designated hardware platform.
- Pre-recorded videos may support the demonstration but do not replace a live functional test unless approved by the teaching team.
- The assessor may vary card position, card order, exposure time, lighting or test sequence within reasonable limits.
- Each student may be questioned separately. Team marks are not automatically identical when individual understanding or contribution differs.
- Claimed multicore, RTOS/LINUX, CNN acceleration and hardware–software partitioning features must be supported by technical evidence.
- Source code, FPGA design files, model files and build instructions must be available during assessment.

- The system should recover safely from reset, invalid input or an unsuccessful recognition attempt.

4.1 Viva topics

- Function of each SoC block and the interconnect architecture.
- Data flow from camera to memory, recognition and HDMI output.
- Role of each Arm core and the RTOS/LINUX task structure.
- CNN architecture, dataset, preprocessing, accuracy and FPGA acceleration method.
- Frame buffering, external DRAM access and bandwidth considerations.
- Game-state representation, matching logic, capture logic and recovery logic.
- Testing methodology, limitations and proposed improvements.

5. Two-Page SoC Datasheet Submission

Each team shall submit a concise two-page datasheet describing the implemented Arm Based ECE4813-SoC. The datasheet is a technical product-style summary rather than a conventional long report. It should allow another engineer to understand the system features, interfaces and demonstrated performance quickly.

5.1 Required datasheet content

Page	Required content
Page 1 – Architecture and features	System name and purpose; block diagram; processor and FPGA resources; major hardware accelerators; memory architecture; supported interfaces; operating modes; key software and RTOS/LINUX features.
Page 2 – Performance and application	Camera input and image format; CNN model and classes; recognition accuracy or test results; latency/frame rate where measured; HDMI output features; game modes; resource utilisation; limitations; team identification and version information.

5.2 Datasheet quality expectations

- Maximum length: two A4 pages, excluding an optional source-code repository link or QR code.
- Use concise engineering language, tables, diagrams and measurable specifications.
- Do not claim features that were not implemented or tested.
- Clearly distinguish measured values from estimated or target values.
- Use the official system name: Arm Based ECE4813-SoC.
- Include a revision number and submission date.

6. Final Deliverables

Deliverable	Due/assessment point	Minimum evidence
Start	Week 2	Team Git hub link submitted
Milestone 1 demonstration	Week 6	Functional SoC build, initial recognition and viva.
Milestone 2 demonstration	Week 9	4 × 4 memory game, repeated pair matching and HDMI dashboard.
Milestone 3 demonstration	Week 12	King–Jack–Joker strategy game, captures, scoring and viva.
Two-page SoC datasheet	Week 12	PDF or DOCX datasheet meeting Section 5 requirements.

Deliverable	Due/assessment point	Minimum evidence
Design package	Week 12	Source code, FPGA project, model files, build instructions and test evidence.

7. Summary

The mini-project integrates Arm multicore software, FPGA-based acceleration, embedded vision, memory systems, peripheral interfaces and interactive game logic. Teams are expected to evolve their Arm Based ECE4813-SoC progressively: first establishing a functioning architecture, then demonstrating reliable card recognition and matching, and finally implementing an advanced competitive strategy game with autonomous state tracking and HDMI visualisation.